

JPHG02 Heat and internal energy — Quiz 2

S3 Physics · 3 multiple-choice items

JPHG02.4 Question 1

Jimmy uses a joulemeter to find the specific heat capacity of water.

- Mass of water = 0.2 kg
- Initial temperature = 25 °C
- Final temperature = 42 °C
- Initial joulemeter reading = 64 350 J
- Final joulemeter reading = 79 470 J

What is the specific heat capacity of water?

- **A.** 1 800 J kg⁻¹ °C⁻¹
- **B.** 3 020 J kg⁻¹ °C⁻¹
- **C.** 4 450 J kg⁻¹ °C⁻¹
- **D.** 23 400 J kg⁻¹ °C⁻¹

JPHG02.5 Question 2

A heater supplies 1000 J of energy to a 5 kg metal block. The temperature of the metal block rises from 25 °C to 35 °C. Assume that there is no energy loss to the surroundings. What is the heat capacity of the metal block?

- **A.** 20 J °C⁻¹
- **B.** 50 J °C⁻¹
- **C.** 100 J °C⁻¹
- **D.** 200 J °C⁻¹

JPHG02.6 Question 3

A metal block of mass 0.1 kg at 150 °C is put in water of mass 1 kg at 20 °C. The heat capacity of the metal block is 100 J °C⁻¹. What is the final temperature of the mixture?

Specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹

- **A.** 20.3 °C
- **B.** 23.0 °C
- **C.** 85 °C
- **D.** Cannot be determined

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Teacher page · 3 items

JPHG02.4 Question 1

Answer: C. $4\,450\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

Guide: Energy supplied = $79\,470 - 64\,350 = 15\,120\text{ J}$. $\Delta T = 42\text{ }^{\circ}\text{C} - 25\text{ }^{\circ}\text{C} = 17\text{ }^{\circ}\text{C}$. $c = 15\,120 / (0.2 \times 17) = 4\,450\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$.

JPHG02.5 Question 2

Answer: C. $100\text{ J }^{\circ}\text{C}^{-1}$

Guide: $C = Q / \Delta T = 1000\text{ J} / (35\text{ }^{\circ}\text{C} - 25\text{ }^{\circ}\text{C}) = 100\text{ J }^{\circ}\text{C}^{-1}$.

JPHG02.6 Question 3

Answer: B. $23.0\text{ }^{\circ}\text{C}$

Guide: Energy lost by the block equals energy gained by the water: $100 \times (150 - T) = 1 \times 4200 \times (T - 20)$, so $T = 23.0\text{ }^{\circ}\text{C}$.